

Amey Agrawal

CS Ph.D. Student, Georgia Tech

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Education

May 2026 Aug 2022	Georgia Institute of Technology Ph.D. Candidate, Computer Science. GPA 4.00/4.00. Advisor: Prof. Alexey Tumanov / Area: Systems for machine learning, LLM inference systems.	Atlanta, USA
Jul 2018 Aug 2014	Birla Institute of Technology and Science Pilani B.E. (Hons.), Computer Science	Pilani, India

Experience

Present Jan 2025	Project Vajra Project Lead / Mentor: Prof. Alexey Tumanov Leading a team comprised of 14 master's students and 3 PhD students to build an LLM serving system from scratch with native real-time multi-modal support.	Atlanta, USA
Present May 2025	Microsoft Research Research Intern / Mentor: Dr. Sadjad Fouladi , Dr. Ganesh Ananthanarayanan High-performance distributed inference systems for large language models.	Redmond, USA
Aug 2024 May 2024	Microsoft Azure Systems Research Research Intern / Mentor: Dr. Esha Choukse Built systems to serve large language models with multimillion context length requests.	Redmond, USA
Aug 2023 May 2023	Microsoft Research Research Intern / Mentors: Dr. Ramchandran Ramjee , Dr. Bhargav Gulavani Designed efficient inference systems for large language models.	Bangalore, India
Aug 2022 Jan 2021	Microsoft Research Research Software Engineer-II / Mentor: Dr. Muthian Sivathanu Built parts of the elasticity sub-system that leveraged efficient time sharing of GPUs to provide transparent scaling of deep learning training workloads. This work was done as a part of the Singularity project, Microsoft's planet-scale AI infrastructure service.	Bangalore, India
Nov 2020 Jul 2018	Qubole Inc. Member of Technical Staff-II / Mentor: Rohit Karlupia Worked on various applied machine learning and software engineering problems to enhance Qubole's data science platform. Published research in several top-tier venues.	Bangalore, India
Dec 2017 Jul 2017	Software Engineering Intern / Mentor: Bharath Bhushan Built core data-plane components for Qubole's Deep Learning clusters based on TensorFlow and Apache Spark.	

Publications

Akasha: Breaking the I/O Wall in Hierarchical KV Cache Management

Amey Agrawal*, Elton Pinto*, Souradeep Bera*, Sukrit Kumar, Irene Paul Stephen, Chus Antonanzas, Anirudha Agrawal, Aldinash Seitenov, Prakhar Jagwani, Anmol Agarwal, Haoran Qiu, Alexey Tumanov
Preprint

Setu: A Global Tensor Exchange for Composable Transfers in Large-Scale ML Systems

Elton Pinto*, Amey Agrawal*, Aldinash Seitenov, Anirudha Agrawal, Bhumika Chopra, Roshan Dathathri, Sadjad Fouladi, Alexey Tumanov
Preprint

Revati: Transparent GPU-Free Time-Warp Emulation for LLM Serving [pdf]

Amey Agrawal*, Mayank Yadav*, Sukrit Kumar, Anirudha Agrawal, Garv Ghai, Souradeep Bera, Elton Pinto, Sirish Gambhira, Mohammad Adain, Kasra Sohrab, Chus Antonanzas, Alexey Tumanov
Preprint: [arXiv:2601.00397](https://arxiv.org/abs/2601.00397) (2026)

[CoRR]

- Maya: Optimizing Deep Learning Training Workloads using GPU Runtime Emulation** [\[pdf\]](#)
 Srihas Yarlagadda*, **Amey Agrawal***, Elton Pinto*, Hakesh Darapaneni, Mitali Meratwal,
 Pranavi Bajjuri, Shivam Mittal, Srinivas Sridharan, Alexey Tumanov
Proceedings of 21st European Conference on Computer Systems, 2026, Edinburgh [EuroSys'26]
- No Request Left Behind: Tackling Heterogeneity in Long-Context LLM Inference with Medha** [\[pdf\]](#)
Amey Agrawal, Haoran Qiu, Junda Chen, Íñigo Goiri, Chaojie Zhang, Rayyan Shahid,
 Ramachandran Ramjee, Alexey Tumanov, Esha Choukse
Preprint: arXiv:2409.17264 (2024) [CoRR]
- Taming Throughput-Latency Tradeoff in LLM Inference with Sarathi-Serve** [\[pdf\]](#)[\[code\]](#)[\[video\]](#)
Amey Agrawal, Nitin Kedia, Ashish Panwar, Jayashree Mohan, Nipun Kwatra, Bhargav S. Gulavani,
 Alexey Tumanov, Ramachandran Ramjee
Proceedings of 18th USENIX Symposium on Operating Systems Design and Implementation, 2024, Santa Clara [OSDI'24]
- Vidur: A Large Scale Simulation Framework For LLM Inference** [\[pdf\]](#)[\[code\]](#)[\[video\]](#)
Amey Agrawal, Nitin Kedia, Jayashree Mohan, Ashish Panwar, Nipun Kwatra, Bhargav S. Gulavani,
 Ramachandran Ramjee, Alexey Tumanov
Proceedings of 7th Annual Conference on Machine Learning Systems, 2024, Santa Clara [MLSys'24]
- On Evaluating Performance Of LLM Inference Serving Systems** [\[pdf\]](#)
Amey Agrawal, Nitin Kedia, Anmol Agarwal, Jayashree Mohan, Souvik Kundu, Nipun Kwatra,
 Ramachandran Ramjee, Alexey Tumanov
Preprint: arXiv:2507.09019 (2025) [CoRR]
- Inshrinkerator: Compressing Deep Learning Training Checkpoints via Dynamic Quantization** [\[pdf\]](#)
Amey Agrawal, Sameer Reddy, Satwik Bhattamishra, Sarath Nookala,
 Vidushi Vashishth, Kexin Rong, and Alexey Tumanov
Proceedings of 15th ACM Symposium on Cloud Computing, 2024, Redmond [SoCC'24]
- Sarathi: Efficient LLM Inference by Piggybacking Decodes with Chunked Prefills** [\[pdf\]](#)
Amey Agrawal, Ashish Panwar, Jayashree Mohan, Nipun Kwatra, Bhargav S. Gulavani, Ramachandran Ramjee
Preprint: arXiv:2308.16369 (2023) [CoRR]
- Sybill: Deep Learning Workload Tuning with Virtual GPUs** [\[poster\]](#)
 Srihas Yarlagadda*, **Amey Agrawal***, Sarath Nookala, Pranavi Bajjuri, Shivam Mittal, Alexey Tumanov
ACM Symposium on Cloud Computing Poster, 2023 [SoCC'23]
- Singularity: Planet-Scale, Preemptible, Elastic Scheduling of AI Workloads** [\[pdf\]](#)
 Singularity Team, Microsoft
Preprint: arXiv:2202.07848 (2022) [CoRR]
- Logan: A Distributed Online Log Parser** [\[pdf\]](#)
Amey Agrawal, Rajat Gupta, and Rohit Karlupia
Proceedings of IEEE International Conference on Data Engineering, 2019, Macau [ICDE'19]
- Learning Digital Circuits: A Journey Through Weight Invariant Self-Pruning Neural Networks** [\[pdf\]](#)[\[code\]](#)
Amey Agrawal, and Rohit Karlupia
Sparsity in Neural Networks Workshop 2021; New in ML Workshop, NeurIPS, 2019, Vancouver [SNN'21]
- Delog: A Privacy Preserving Log Filtering Framework for Online Compute Platforms** [\[pdf\]](#) [\[dataset\]](#)
Amey Agrawal, Abhishek Dixit, Namrata Shettar, Darshil Kapadia,
 Rohit Karlupia, Vikram Agrawal, and Rajat Gupta
Proceedings of IEEE International Conference on Big Data, 2019, Los Angeles [BigData'19]

Honours and Awards

Center for Research into Novel Compute Hierarchies (CRNCH) Fellowship, 2023 [\[🌐\]](#)

> For research of automatic hardware-aware optimization of deep learning training workloads.

School of Computer Science Fellowship, 2022

> PhD fellowship from Georgia Institute of Technology.

Teaching Roles

Systems for Machine Learning *Head Teaching Assistant / Georgia Institute of Technology*

Fall'24

- › Conducted series of lectures on GPU architecture and LLM inference.

Introduction to Neural Networks & Fuzzy Logic *Head Teaching Assistant* [\[assignments\]](#)

Aug'17 - May'18

- › Introduced Python programming assignments along with a new custom-built evaluation platform. Other responsibilities included coordinating the team of seven teaching assistants to conduct labs, designing assignments and helping students with the term project.

Introduction to Machine Learning *Teaching Assistant*

Jan'18 - May'18

- › Conducted introductory sessions on the scientific Python ecosystem, and organized tests and programming assignments for over 100 students in the class.